Task 2 – Semantics segmentation and depth estimation

2.1 Dataset Description and Visualization

The Cityscapes dense prediction dataset used in this task contains 800 training and 250 validation samples, each consisting of a 128×256 RGB image along with two corresponding ground truth maps: a semantic segmentation mask (19 classes) and a depth map. The depth maps are derived from disparity data using the formula depth = (B × f) / d, where B = 0.222384 is the camera baseline and f = 2273.82 is the focal length. Invalid depth values (zero or negative) are masked out during training and evaluation. Some samples with their corresponding semantic and depth maps are illustrated in Figure 3.

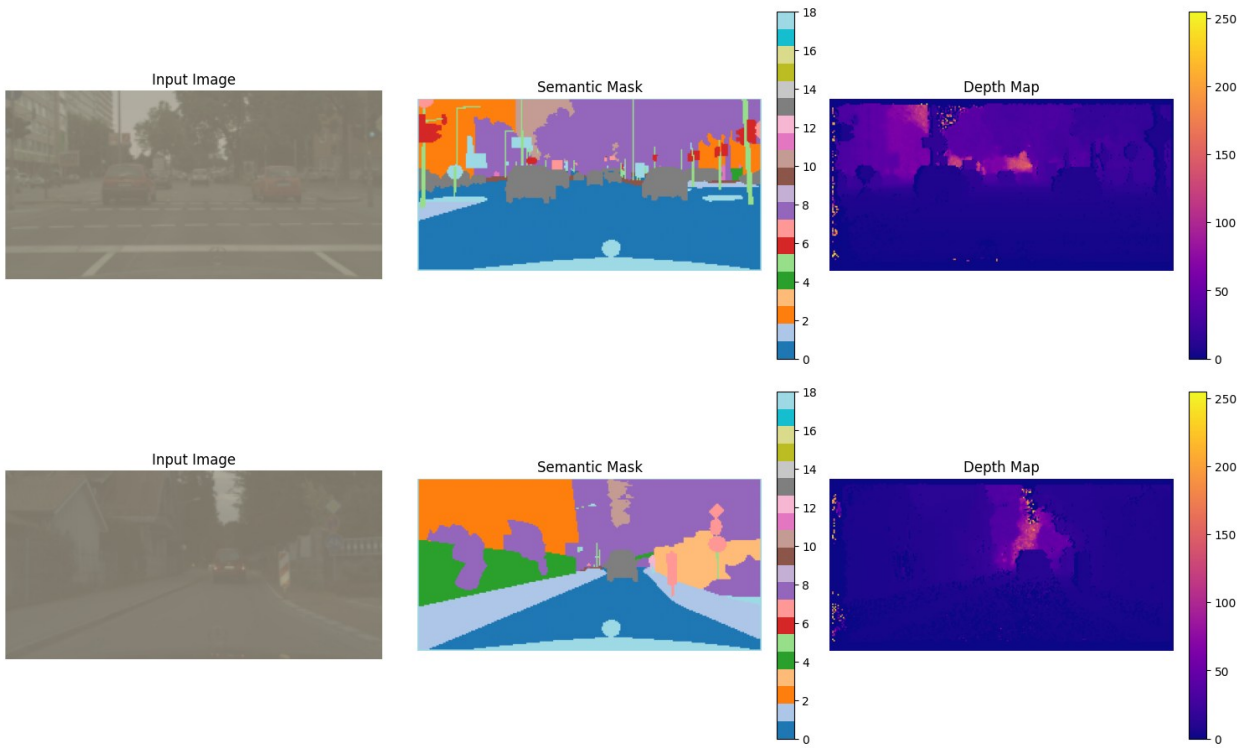

Figure 3: Example training samples with semantic and depth ground truth.

The dataset exhibits significant class imbalance, with classes like "road" and "building" appearing frequently, while classes such as "train" and "motorcycle" are rare. To address this, class weights provided in *classweight.cls* are applied to the cross-entropy loss during training.

2.2 Single-Task FCN for Semantic Segmentation (FCN_ST)

The single-task FCN uses a ResNet50 backbone pre-trained on ImageNet as an encoder, followed by a decoder with skip connections to recover spatial resolution. The architecture is fully convolutional, allowing it to handle arbitrary input sizes while producing output maps of the same spatial dimensions. Transposed convolutions and output padding are used for upsampling, and skip connections combine high-level semantic features from the encoder with low-level spatial details from earlier layers to produce sharp segmentation boundaries. Table 3 presents the performance of the single-task model on both training and validation sets.

Table 3: Single-Task FCN Performance.

Single-Task	Segmentation (Accuracy %)	mIoU(%)
Training	87.78	62.40
Validation	82.95	37.27
Download link	fcn_st.pth	

The mean Intersection over Union (mIoU) is the primary metric for segmentation tasks. For a single class, IoU is defined as:

$$IoU = (True\ Positives) / (True\ Positives + False\ Positives + False\ Negatives)$$

mIoU averages this value across all 19 classes, providing a balanced measure that penalizes both false positives and false negatives. Unlike pixel accuracy, which can be misleading in imbalanced datasets, mIoU gives equal weight to all classes, making it the standard evaluation metric for segmentation tasks.

Table 4 shows the per-class IoU for the validation set, revealing that classes like "road" (92.79%) and "sky" (84.60%) achieve high scores, while rare classes such as "motorcycle" (6.32%) and "train" (14.97%) perform poorly due to limited training examples.

Table 4: Per-Class IoU for Single-Task FCN (Validation Set).

Class	IoU (%)	Class	IoU (%)
0 road	92.79	9 terrain	41.43
1 sidewalk	60.38	10 sky	84.60
2 building	69.75	11 person	32.41
3 wall	12.71	12 rider	13.82
4 fence	12.01	13 car	77.26
5 pole	15.67	14 truck	13.29
6 traffic light	11.33	15 bus	29.20
7 traffic sign	18.04	16 train	14.97
8 vegetation	72.70	17 motorcycle	6.32
		18 bicycle	29.38

2.3 Multi-Task FCN (FCN_MT)

Building upon the single-task architecture, we introduce a multi-task version that simultaneously predicts semantic segmentation and depth estimation. The model shares the ResNet50 encoder but uses two separate decoder branches, one for segmentation and one for depth, each with its own skip connections. The depth branch outputs a single-channel map scaled to represent depth in meters.

Training follows a two-stage process:

1. **Pretraining**: The single-task segmentation model (FCN_ST) is trained first.
2. **Fine-tuning**: The multi-task model is initialized with the single-task weights (excluding the depth head) and then jointly optimized using a combined loss:
    - **Segmentation Loss**: Cross-entropy with class weights (same as single-task)
    - **Depth Loss**: L1 loss computed only on valid depth pixels ($gt > 0$)
    - **Total Loss**: $L_total = L_seg + \lambda \cdot L_depth, with\ \lambda = 1.0$

Table 5: Metric results for multitask FCN_MT.

Multi-Task	**Segmentation**		**Depth**			
	Accuracy(%)	mIoU(%)	Rel	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Training	88.04	62.52	0.0917	89.57	95.58	97.12
Validation	82.07	37.19	0.1899	72.51	89.22	94.45
Download link			fcn_mt.pth			

The depth metrics are defined as listed in table 6:

Table 6: metric explanation.

Metric	**Formula**	**Interpretation**
rel (Mean Relative Error)	$\left(\frac{1}{N}\right) \Sigma \frac{\lvert pred - gt \rvert}{gt}$	Average percentage error. Lower is better (<0.15 is good).
$\delta < t$	$\max\left(\frac{pred}{gt}, \frac{gt}{pred}\right) < t$	Percentage of pixels within a given threshold. Higher is better.

The thresholds used are:

- $\delta < 1.25$: Prediction within 25% of true depth
- $\delta < 1.25^2$ (≈1.56): Within 56% error
- $\delta < 1.25^3$ (≈1.95): Within 95% error

The validation results show that depth estimation is reasonably accurate, with 72.51% of pixels within 25% error and a mean relative error of 18.99%. The per-class IoU for segmentation (Table 7) remains comparable to the single-task variant.

Table 7: Per-Class IoU for Multi-Task FCN (Validation Set).

Class	IoU (%)	Class	IoU (%)
0 road	91.81	9 terrain	37.20
1 sidewalk	57.10	10 sky	84.95
2 building	69.16	11 person	30.86
3 wall	8.77	12 rider	12.50
4 fence	11.10	13 car	78.30
5 pole	15.54	14 truck	10.66
6 traffic light	11.88	15 bus	36.59
7 traffic sign	17.83	16 train	24.49
8 vegetation	71.92	17 motorcycle	7.11
		18 bicycle	28.75

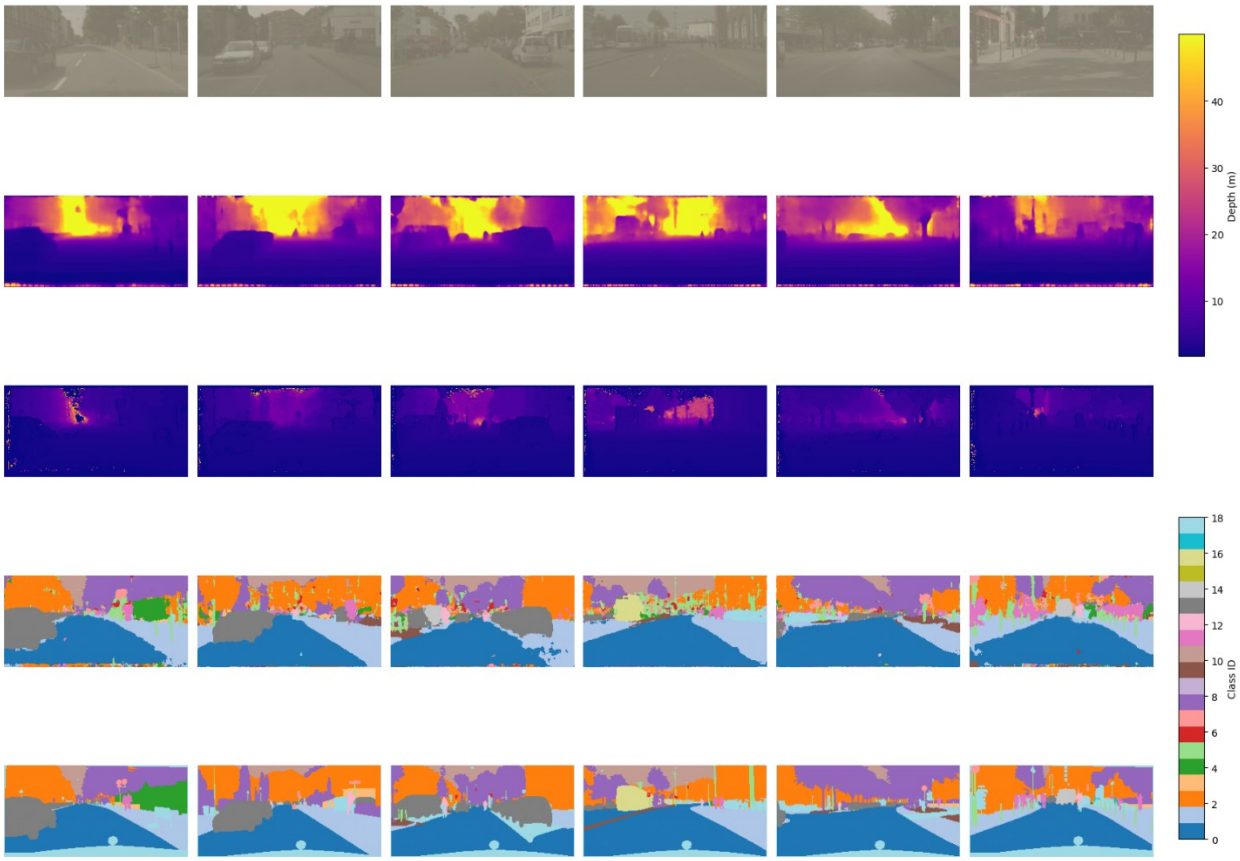

Figure 4: Results of the trained model for semantic segmentation and depth estimation.

Some sample results are illustrated in Figure 4. First row contains the input images, Row 2 presents the predicted depth maps, Row 3 provides the ground truth depth maps, Row 4 displays the predicted semantic maps, and Row 5 includes the ground truth semantic maps.

2.4 Cross-Dataset Evaluation on KITTI

To assess generalization, the multi-task model trained on Cityscapes is evaluated on 6 sample images from the KITTI dataset, which features different camera perspectives (car-mounted vs. street view), scene compositions (more road and car pixels, fewer buildings), and lighting conditions. Since KITTI test samples lack ground truth annotations, only qualitative results are reported. Figure 5 shows the model's predictions on KITTI images:

- **Row 1**: Input KITTI images
- **Row 2**: Predicted depth maps
- **Row 3**: Predicted semantic segmentation.

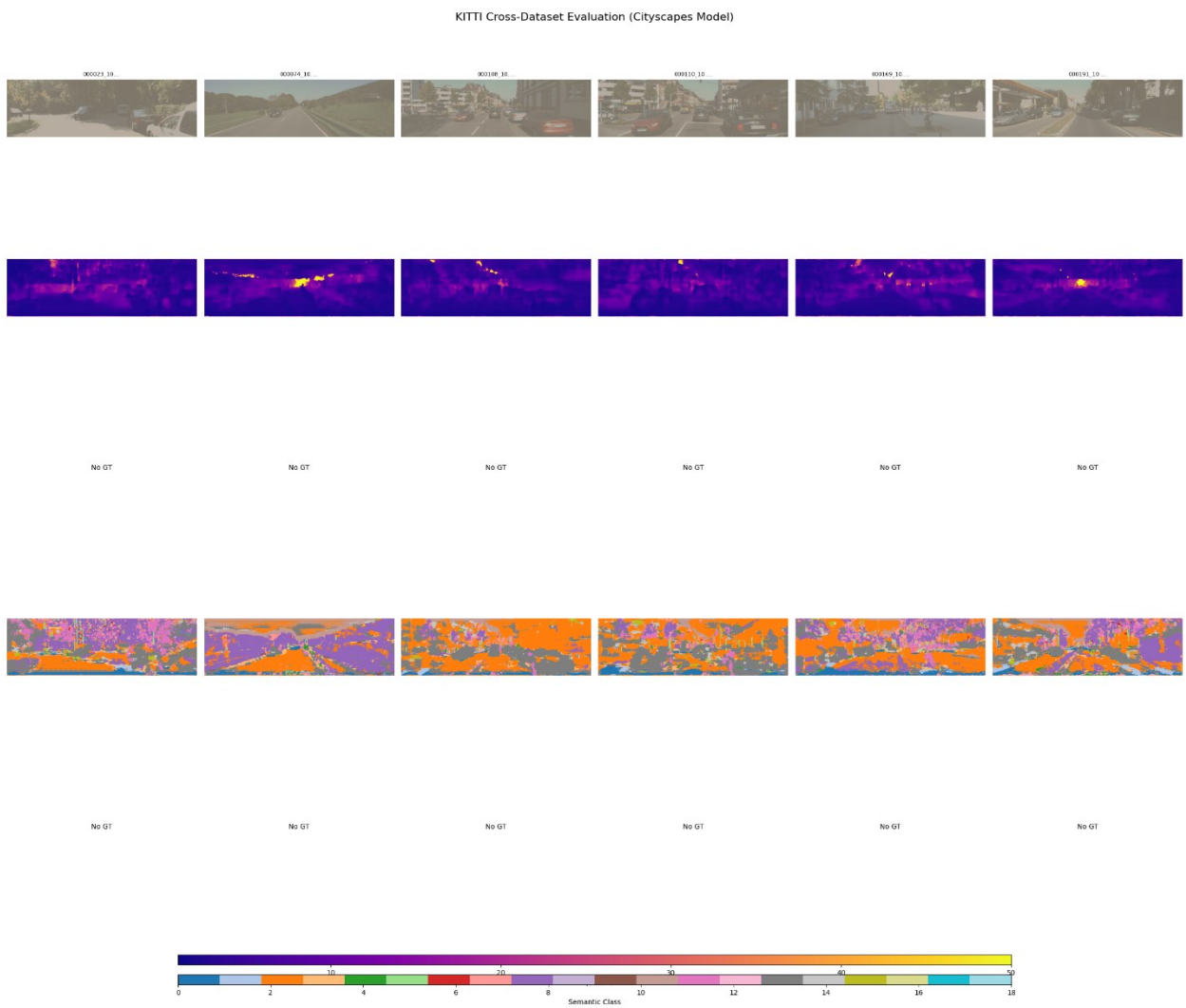

Figure 5: KITTI cross-dataset evaluation results.